



## AJKD Atlas of Renal Pathology: Nephrocalcinosis and Acute Phosphate Nephropathy

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### Clinical and Pathologic Features

Nephrocalcinosis, the deposition of calcium phosphate crystals within renal tubules and sometimes within interstitium, may occur at any age. Patients may present with acute kidney injury (AKI). In patients with acute phosphate nephropathy, progression to end-stage kidney disease (ESKD) can occur, particularly if there is pre-existing chronic kidney disease (CKD). Acute phosphate nephropathy typically occurs due to a large load of oral sodium phosphate as part of colonoscopy preparation, and presents as AKI, developing in 1 in 1,000 to 5,000 of such patients. Risk factors include dehydration, pre-existing CKD, concomitant use of ACE inhibitors or diuretics, older age, and female sex. Recognition of this entity and change in colonoscopy preparation have markedly decreased incidence of acute phosphate nephropathy. Nephrocalcinosis can also be due to excess calcium, due to either hypercalcemia or hypercalciuria, and may occur at any age. The prognosis depends upon the underlying cause. CKD and even ESKD can develop if deposits are extensive and the underlying cause cannot be treated. In patients with acute phosphate nephropathy, serum calcium levels are typically normal, in contrast to nephrocalcinosis caused by hypercalcemia.

Light microscopy: In the acute phase, there is acute tubular injury with calcium phosphate crystals, with granular chunky, bluish-purplish nonpolarizing deposits of calcium phosphate within tubular lumens. Acute tubular injury is manifest by sloughing of tubular epithelial cells, vacuolization, blebbing, and simplified tubular epithelium. Phosphate and calcium within these deposits are detectable with von Kossa

and alizarin stains, respectively. In the chronic phase, fewer calcium phosphate crystals are present, with lesser acute tubular injury and predominant tubular atrophy and interstitial fibrosis, with mild interstitial lymphoplasmacytic infiltrate. Occasional calcium phosphate deposits may also be seen in the interstitium. Calcium phosphate deposition can also rarely occur along tubular basement membranes and even more rarely in glomeruli in the mesangium or along the glomerular basement membranes.

Immunofluorescence microscopy: No staining.

Electron microscopy: No specific changes.

### Etiology/Pathogenesis

Any condition with hypercalcemia, hypercalciuria, hyperphosphatemia, or hyperphosphaturia may result in calcium phosphate precipitates within renal tubules whenever the solubility coefficient is exceeded. Hypercalcemic conditions include sarcoidosis, malignancy, hyperparathyroidism, vitamin D therapy, milk-alkali syndrome, and congenital hypothyroidism. Hypercalciuria can be seen in medullary sponge kidney, and rare inherited diseases such as Dent disease, cystinosis, and Bartter syndrome. Some drugs (eg, loop diuretics) can cause hypercalciuria. Hypocitraturia, especially in patients with type 1 renal tubular acidosis, may also contribute by inducing hypercalciuria. Acute loads of phosphate, as with sodium phosphate ingestion for colonoscopy, may also cause nephrocalcinosis.

### Differential Diagnosis

Calcium phosphate crystals are bluish-purple and nonpolarizable. Other crystals, such as calcium oxalate or 2,8-dehydroxyadenine, polarize. Of note, nonspecific occasional calcium phosphate crystals may be seen secondary to any CKD and are not specifically indicative of an underlying calcium or phosphate abnormality.

### Key Diagnostic Features

- Acute phase: acute tubular injury with numerous bluish-purplish intratubular nonpolarizable calcium phosphate crystals
- Chronic phase: interstitial fibrosis and tubular atrophy with calcium phosphate crystals

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Support: None.

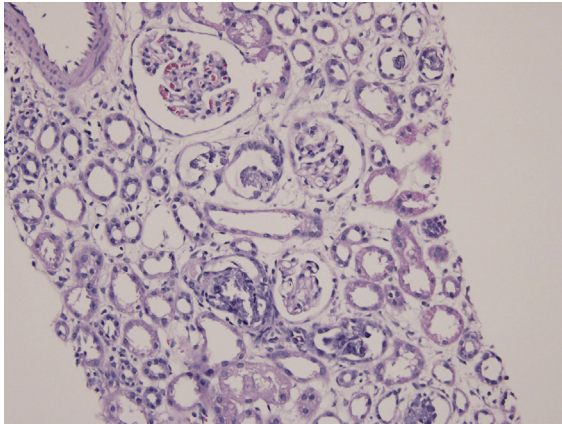
Financial Disclosure: The authors declare that they have no relevant financial interests.

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*Am J Kidney Dis.* 69(3):e17-e18.

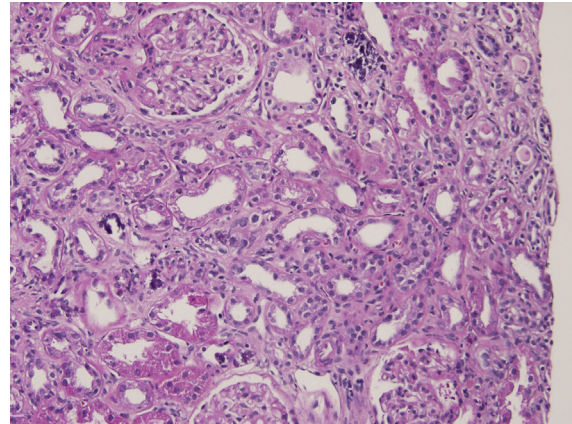
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0272-6386

<http://dx.doi.org/10.1053/j.ajkd.2017.01.005>



**Figure 1.** Nephrocalcinosis and acute phosphate nephropathy with intratubular calcium phosphate crystals and acute tubular injury (hematoxylin and eosin stain).



**Figure 2.** Nephrocalcinosis and acute phosphate nephropathy with intratubular calcium phosphate crystals, tubular atrophy and interstitial fibrosis (hematoxylin and eosin stain).